

Emergent Process Design

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1. RESEARCH PROBLEM

The typical situation in software developing organizations can be characterized by two extremes: in some organizations or organizational units, software development processes are partly defined, whereas in organizations or organizational units where the processes are well defined, these processes are usually only valid at the project start. This is the case as the development and business environments change over the course of the project. The environment changes are caused, for instance, by new innovations, new business relationships, and new technical and cost constraints.

In both cases (i.e., defined processes only or out-dated processes), the likely consequences vary for those involved: for all project participants, lack of coordination; for developers, fire-fighting and overtime; for project managers, deteriorating progress control; for the customer, higher costs to achieve the required product quality. Hence, practitioners need support in the design and evolution of project-specific processes over the course of a project in a dynamic project environment.

2. PROPOSED SOLUTION

2.1 Requirements

In order to understand how a software development process should look if it is to be acceptable by both project managers and engineers, I first defined the requirements placed on the processes. To do so, I held workshops and interviews with DaimlerChrysler project managers, engineers, and experts in the domain of process design and adaptation. This helped me to understand the strengths and deficiencies of applied software development processes from the interviewees' and workshop participants' points of view. Then I monitored the development processes in different DaimlerChrysler projects and analyzed them in order to understand the real cause of the process deficiencies. Second, I defined the possible process flexibility levels and the necessary process flexibility level for the particular issue of development of large, safety critical software in a dynamic environment. Third, the actual definition of the demands placed for processes was geared at the GQM method for the required process flexibility level. Finally, the importance of the requirements on the processes was validated [2].

2.2 Concept

For design of a process that is able to fulfill the requirements, I propose an Emergent Process Design (EPD) approach. The EPD approach guides the extraction and the adaptation of a project-specific process. This is done as follows. Initially, the project-specific process frame is extracted by tailoring one of the process variants provided in a process line. The first development is based on the tailored process frame. For further development iterations or projects that are to utilize this frame, the frame is adapted and refined to the project-specific process based on the logs of the performed tasks in the previous process iteration. In order to insure the desired quality of the final product, the project-specific process becomes more stringent over the course of the project.

As the approach should also be suitable for a dynamic environment, the EPD approach should guide the adaptation of the prescriptive process to the emerging project situations. The systematic process adaptation is insured by the definition of modification preconditions. These modification preconditions describe the criteria to be fulfilled before a process can be modified. Thus, the process can easily and systematically be adapted to new situations by the application of a suitable process variant.

2.3 The Empirical Validation

The EPD approach is currently being validated in the context of a controlled experiment. In the experiment, we compare the EPD method with the V method [4]. The V model is taken for comparison, as it is a widely used method in practice. In the experiment, nine groups (six following the EPD approach and three following the V tailored process) develop a control device for a bus. The first results show that the EPD method significantly contributes to a more productive development of higher-quality products.

3. RELATED WORK

The topic addressed in my thesis has been discussed by Jorgensen [3] and Cugola [1].

4. ACKNOWLEDGMENTS

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5. REFERENCES

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